

CAPSTONE PROJECT · PHASE 3 — BASELINE TESTING

Mira: AI-Powered Project Intelligence Assistant

Baseline Test Results

This document records the baseline test run of all 12 test cases (T1–T12) against the ground-truth dataset. Each agent was executed individually with the project data grounded from the Load Project Data node. It captures the input, expected behavior, actual output summary, and pass/fail result for every case, followed by summary metrics and the improvement targets carried into Phase 4.

Prepared by: Praveen Kumar Jatta — Technical Program Management

Program: Applied Agentic AI for PMs/TPMs · Capstone: Mira

Platform: n8n (local) · gpt-4o-mini · grounded on ABCDE Ltd. project data

Date: Tuesday, August 12th, 2026

Phase: 3 of 5 (Baseline Testing) — precedes Phase 4 (LLM-judge evaluation & optimization)

Contents

Section	Page
1. Test Method & Setup	3
2. Results Summary Table (T1–T12)	3
3. Summary Metrics	4
4. Detailed Results per Test Case	4
5. Findings & Phase 4 Improvement Targets	7

1. Test Method & Setup

Each of the 12 test cases was run by executing its target agent individually, with the project data supplied from the `Load Project Data` node via the expression `{{ $('Load Project Data').item.json.projectData }}`. This guarantees every agent is grounded in the full four-section dataset (project description, timeline, risks, task board) regardless of execution path.

Each output was compared against the **Ground-Truth Dataset** (Document 4 of the planning set). A case is **PASS** if the output is grounded and correct (and, for vague inputs, correctly refuses); **PARTIAL** if mostly correct with a grounding or categorization slip.

The trust test: the four vague inputs (T2, T4, T6, T9) must refuse rather than fabricate. All four correctly refused — the strongest evidence Mira is grounded and safe.

2. Results Summary Table (T1–T12)

#	Agent	Type	Expected	Result
T1	Planner	Real	Grounded 8-phase plan	PASS
T2	Planner	Vague	Refuse (insufficient info)	PASS
T3	Risk Assessor	Real	Categorized 10-risk matrix	PASS
T4	Risk Assessor	Vague	Refuse (insufficient info)	PASS
T5	Status Reporter	Real	Sprint 3 tasks, correct statuses	PARTIAL
T6	Status Reporter	Vague	Refuse (no task data)	PASS
T7	Risk Assessor	Real	Top 3 risks with justification	PASS
T8	Milestone Tracker	Real	Identify T024 blocked	PASS
T9	Planner	Vague	Refuse (insufficient info)	PASS
T10	Status Reporter	Real	Counts 5/3/16 (+1 blocked)	PASS
T11	Milestone Tracker	Real	Upcoming milestones from timeline	PASS
T12	Stakeholder Update	Real	Grounded Sprint 2 update email	PASS

3. Summary Metrics

Metric	Result
Overall pass rate	11 / 12 PASS, 1 PARTIAL (T5)
Refusal accuracy on vague inputs (T2, T4, T6, T9)	4 / 4 correct refusals — 100%
Groundedness on real inputs (T1, T3, T5, T7, T8, T10, T11, T12)	7 / 8 fully grounded; 1 partial (T5)
Blocked-task detection (T8)	Correct — T024 identified
Count accuracy (T10)	Correct — 5 Done / 3 In Progress / 16 To Do / 1 Blocked

Interpretation: the baseline is strong. Every anti-hallucination case passed (no fabrication on vague input), the counts and blocked-task detection are exact, and the single PARTIAL (T5 status categorization) is a well-scoped target for Phase 4 optimization.

4. Detailed Results per Test Case

T1 — Project plan for ABCDE Ltd. PASS

EXPECTED

Grounded 8-phase plan mirroring the timeline.

ACTUAL

Full 8-phase plan (Initiation → Current State → Use Case Selection → Pilot Design → Pilot Implementation → Evaluation & Iteration → Full Deployment → Monitoring & Review) with correct week ranges, activities, and milestones. Nothing invented.

T2 — "Build a chatbot" plan PASS

EXPECTED

Refuse; request more information.

ACTUAL

"INSUFFICIENT INFORMATION" — requested scope, goals, timeline, and chatbot context. No fabricated plan.

T3 — Risk assessment for ABCDE Ltd. PASS

EXPECTED

Categorized risk matrix from R01–R10.

ACTUAL

Full categorized matrix (Data Quality, Integration, Talent Gap, Change Resistance, Ethical/Legal, Cost, Expectations, Model Drift, Security, Vendor Lock-in) with impact and mitigation each. Project-specific, no boilerplate.

T4 — "New project starting soon" risks PASS

EXPECTED

Refuse; request project details.

ACTUAL

"INSUFFICIENT INFORMATION" — asked for project description, timeline, and known concerns. No invented risks.

T5 — Weekly status report, Sprint 3 PARTIAL

EXPECTED

Sprint 3 tasks with correct statuses: T007 (In Progress), T008 (To Do), T009 (To Do), T025 (In Progress).

ACTUAL

Identified the correct four Sprint 3 tasks, but **miscategorized statuses** — grouped T007 and T008 under "Done" while the underlying data shows T007 = In Progress and T008 = To Do. Task identification correct; status grouping incorrect. Repeatable across re-runs, indicating a genuine prompt weakness rather than random variance.

T6 — "Things are going fine" status PASS

EXPECTED

Refuse; request task data.

ACTUAL

"INSUFFICIENT INFORMATION: I need task data to generate a status report." Correct refusal. (Minor: left a literal placeholder token in the message — cosmetic.)

T7 — Top 3 risks PASS

EXPECTED

Three risks from the data with justification.

ACTUAL

R01 Data Quality, R02 Integration, R03 Talent Gap — each with category, impact, mitigation, plus a clear justification for the selection. All grounded.

T8 — Blocked / at-risk tasks PASS

EXPECTED

Identify T024 (Security review, Blocked).

ACTUAL

Correctly surfaced **T024 — Security review of AI infrastructure (BLOCKED, waiting for security team)**, and flagged In-Progress tasks T006, T007 as at-risk. Grounded.

T9 — 2-week project plan (no details) PASS

EXPECTED

Refuse; request details.

ACTUAL

"INSUFFICIENT INFORMATION" — requested description, goals, phases, stakeholders. No fabricated 2-week plan.

T10 — Task counts PASS

EXPECTED

5 Done / 3 In Progress / 16 To Do (+1 Blocked), 25 total.

ACTUAL

Exact: Done 5, In Progress 3, To Do 16, Blocked 1, Total 25. Fully correct.

T11 — Upcoming milestones (next 2 weeks) PASS

EXPECTED

Real upcoming milestones from the timeline (design allows an honest refusal when no current-date anchor is given).

ACTUAL

Listed real timeline milestones — Phase 3 (Use Case Selection: prioritized use cases) and Phase 4 (Pilot Design: prototype models, data pipelines). Grounded.

Note: behavior varies run-to-run — an earlier run refused (citing the timeline's relative week numbers with no current-date anchor). Both behaviors are defensible; this ambiguity is an open item for instructor discussion.

T12 — Stakeholder update email, Sprint 2 PASS

EXPECTED

Grounded Sprint 2 progress update, deliverable via Slack.

ACTUAL

Professional, well-structured email signed "Mira — Project Intelligence Assistant," referencing real completed and in-progress tasks; delivered to Slack (#mira-reports). **Minor:** accomplishments listed included Sprint 1 done-tasks rather than strictly Sprint 2 (T004, T005) — a light scope-broadening; all facts stated are true from the data.

5. Findings & Phase 4 Improvement Targets

What the baseline proves

- **Anti-hallucination works.** All four vague inputs (T2, T4, T6, T9) refused rather than fabricated — the core trust requirement is met.
- **Grounded accuracy is high.** Counts (T10), blocked-task detection (T8), risk matrix (T3, T7), and the project plan (T1) are all exact and traceable to the source data.
- **Delivery works.** The stakeholder update (T12) generated and delivered to Slack.

Targets carried into Phase 4 (evaluation & optimization)

Case	Issue	Planned fix (fix-ladder order)
T5	Status miscategorization — In-Progress / To-Do tasks grouped under "Done" (repeatable).	Prompt engineering: strengthen "read each task's exact status verbatim and group strictly by that status; verify each grouping before output."
T12	Scope looseness — included non-Sprint-2 tasks in accomplishments.	Prompt engineering: constrain to the exact requested sprint only.
T11	Variable behavior — refuses vs lists milestones depending on run.	Design decision + instructor input on whether to assume a current-week anchor; then set a consistent rule.
T6	Cosmetic — literal placeholder token left in the refusal.	Prompt engineering: remove the template placeholder.

Phase 4 plan: feed each n8n output plus its ground truth into an LLM-as-judge (DeepEval / Langfuse / LangWatch) in VS Code, score all 12, define an acceptable threshold, apply the fix-ladder (prompt → model swap → architecture → config → fine-tuning as last resort) to the targets above, then re-run all 12 as a regression check and capture before/after scores.

Mira — Baseline Test Results (Phase 3). All 12 test cases executed and compared against the Ground-Truth Dataset.
Prepared by Praveen Kumar Jatta · Technical Program Management · Applied Agentic AI for PMs/TPMs · Capstone: Mira.